

Group Theory: Extra Problem Set

20 Problems for Midsem Preparation

Scope. Twenty problems from Gallian, Chapters 2, 3, 4, and 7 (Groups; Finite Groups and Subgroups; Cyclic Groups; Cosets and Lagrange's Theorem), excluding centers of a group.

Tags. [E] easy, [M] medium (typical exam question), [H] hard.

1 Properties of Groups

(Gallian Ch. 2.)

- 1.1 [E] A list of nine integers is supposed to form a group under multiplication modulo 91. One integer got left out, and the list appears as

$$1, 9, 16, 22, 53, 74, 79, 81.$$

Which integer is missing?

- 1.2 [E–M] For each of the following, determine whether $(S, *)$ is a group. If not, say exactly which axiom fails.

(a) $S = \mathbb{Z}$, $a * b = a - b$

(b) $S = \mathbb{Z}$, $a * b = a + b + ab$

(c) $S = \mathbb{Q} \setminus \{-1\}$, $a * b = a + b + ab$

- 1.3 [M] Show that the set

$$G = \left\{ \begin{pmatrix} a & -b \\ b & a \end{pmatrix} : a, b \in \mathbb{R}, (a, b) \neq (0, 0) \right\}$$

forms a group under matrix multiplication. (This group is secretly isomorphic to a group you already know well. Which one, and why?)

- 1.4 [M] Let a, b be elements of a group G . Prove that ab and ba have exactly the same order (finite or infinite).

- 1.5 [M–H] Let G be a group with the property: for all $x, y, z \in G$, $xy = zx \implies y = z$. Prove that G is abelian.

- 1.6 [M] Prove that $\{1, 2, \dots, n-1\}$ is a group under multiplication modulo n if and only if n is prime.

You may use: for integers a, b there exist $x, y \in \mathbb{Z}$ with $\gcd(a, b) = ax + by$.

2 Finite Groups and Subgroups

(Gallian Ch. 3.)

- 2.1 [E] Show that $H = 2\mathbb{Z}$ (the even integers) is a subgroup of $(\mathbb{Z}, +)$.
- 2.2 [M] Let A, B be subgroups of an *abelian* group G . Let $AB = \{ab : a \in A, b \in B\}$. Prove that AB is a subgroup of G .
- 2.3 [M] Let H and K be subgroups of G , and suppose $x, y \in G$ satisfy $Hx = Ky$. Prove that $H = K$.
- 2.4 [M] Let G be an *abelian* group, and let

$$T = \{g \in G : g \text{ has finite order}\}.$$

Prove that T is a subgroup of G .

3 Cyclic Groups

(Gallian Ch. 4.)

- 3.1 [E] Prove that every cyclic group is abelian.
- 3.2 [E–M] Prove that every subgroup of a cyclic group is cyclic.
- 3.3 [M] Let a, b be elements of a group with $|a| = 10$ and $|b| = 21$. Prove that $\langle a \rangle \cap \langle b \rangle = \{e\}$.
- 3.4 [M–H] Prove that a group has exactly three subgroups if and only if it is cyclic of order p^2 for some prime p .
Hint for the forward direction: call the three subgroups $\{e\}$, H , and G . If $a \notin H$ and $a \neq e$, what must $\langle a \rangle$ be?

4 Cosets and Lagrange's Theorem

(Gallian Ch. 7.)

- 4.1 [E] Let n be a positive integer and $H = n\mathbb{Z} \leq \mathbb{Z}$. Find all the left cosets of H in $\mathbb{Z}$. How many are there?
- 4.2 [M] Prove that every group of order 4 is abelian.
Hint: by Lagrange, every non-identity element has order 2 or 4. If some element has order 4, G is cyclic. If every non-identity element has order 2, expand $(ab)^2 = e$ directly to show $ab = ba$.
- 4.3 [M] Let p be a prime of the form $4k + 3$. Show that $x^2 \equiv -1 \pmod{p}$ has no solution.
Hint: think about the order of a hypothetical solution x inside $U(p)$, and what Lagrange's Theorem says it must divide.
- 4.4 [E–M] Let p be a prime. Solve $x^2 \equiv 1 \pmod{p}$; that is, show the only solutions are $x \equiv \pm 1$.

- 4.5 **[M–H]** (Fermat’s Little Theorem, via Lagrange) Let p be a prime and let a be an integer not divisible by p . By applying Lagrange’s Theorem to the group $U(p)$, prove that

$$a^{p-1} \equiv 1 \pmod{p}.$$

- 4.6 **[H]** Suppose $H \leq G$ has the property that for all $a, b \in G$, $aH = bH \implies Ha = Hb$. Prove that $aHa^{-1} = H$ for every $a \in G$.

20 problems, four topics. Good luck.